FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2015

FULL TEST – I

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 04)** contains 4 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (05 – 08) contains 4 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

   Section-A (09 to 11) contains 1 paragraph having 3 questions. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.

3. **Section-C (01 – 06)** contains 6 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

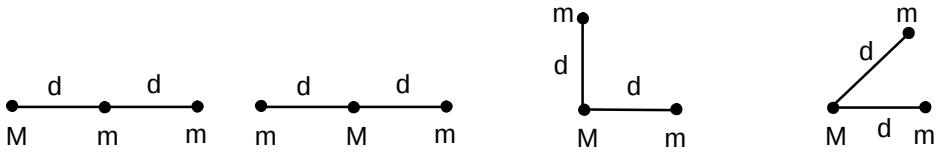
PART – I

SECTION – A

(Only One Option Correct Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

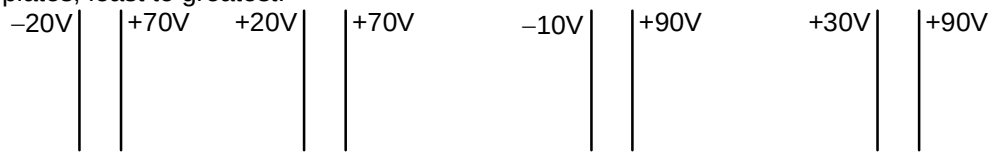
1. Three particles, two with mass m and one with mass M, might be arranged in any of the four configurations shown below. Rank the configurations according to the magnitude of the gravitational force on M, from least to greatest (i.e., in increasing order).



(1) (2) (3) (4)

(A) 1, 2, 3, 4
(B) 2, 1, 3, 4
(C) 2, 1, 4, 3
(D) 2, 3, 4, 2

2. The diagram shows four pairs of large parallel conducting plates. The plates are separated by equal distance in all cases. The value of the electric potential is given for each plate. Rank the pairs according to the magnitude of the electric field between the plates, least to greatest.



(A) 1, 2, 3, 4
(B) 4, 3, 2, 1
(C) 2, 3, 1, 4
(D) 2, 4, 1, 3

Space for Rough work

3. An observer 'O' at an accelerated trolley observes a projectile Q of mass 5 kg moving under uniform gravitational field of earth acting towards negative z-axis. He observes the momentum $\vec{P}$ of the projectile Q as $\vec{P} = 15t\hat{i} - 10t^2\hat{j} - 50t\hat{k}$ (kg m/s), where t denotes time in second. Find the acceleration of the trolley at t = 2 sec. (Take g = 10 m/s^2)

(A) $6\hat{i} - 8\hat{j} - 20\hat{k}$ m/s^2 (B) $3\hat{i} - 8\hat{j}$ m/s^2

(C) $-3\hat{i} - 8\hat{j}$ m/s^2 (D) $-3\hat{i} + 8\hat{j}$ m/s^2

4. A plane thick wall having uniform surface temperature along planes are 0°K and T_0°K (T_0 = 300K) at x = 0 and x = x_0 respectively. Thermal conductivity varies linearly with temperature
K = K_0 (1 + T) The temperature of wall at the plane x = $2x_0$ is approximately: (where T is in kelvin)

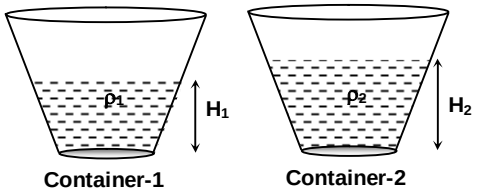
(A) 300K (B) 400K

(C) 425K (D) 450K

(One or More than One Options Correct Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

5. Two identical containers contains liquid of same mass m but different densities as shown in the figure given $\rho_1 > \rho_2$. P_1 and P_2 are pressures at the bottoms of the container-1 and container -2 respectively and F_1 and F_2 are the forces exerted by the liquid on the walls of container-1 and container -2 respectively. (Neglect the atmospheric pressure)

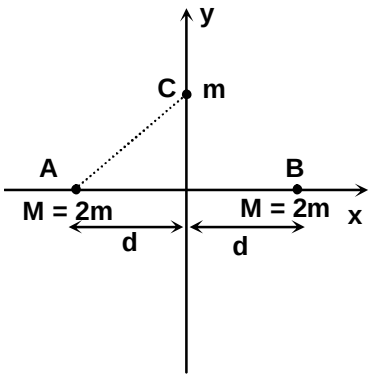


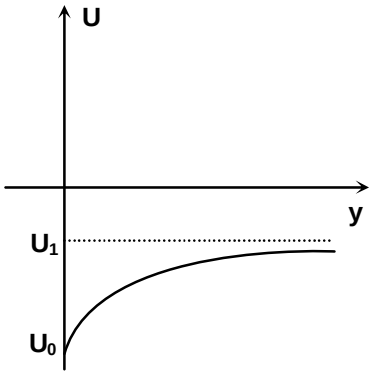
(A) $P_1 > P_2$ (B) $P_1 < P_2$

(C) $F_1 > F_2$ (D) $F_1 < F_2$

Space for Rough work

6. Two point masses, each of mass M are kept at rest at points A and B respectively. A third point m is released from infinity with a negligible speed, so that it can move along y-axis under the influence of mutual gravitational attraction on it due to point masses kept at A and B respectively as shown in the figure -1. Figure –2 represents the potential energy of system (includes m, M at A and M at B) with position of m at y-axis. (Neglect any other forces other than gravity) (given $Gm^2/d = 12$ Joule(m = 6 kg). Choose the correct option(s)





(A) Point mass m will perform periodic motion (B) $U_1 = -24$ Joule
(C) Maximum speed of particle is 24 m/s (D) Maximum speed of particle is 4 m/s

7. Choose the correct statements:
(A) The de broglie wavelength of electron, which are accelerated by an electric potential V is $\lambda = \frac{h}{\sqrt{2m_e eV}}$
(B) Wave velocity of the matter field is one half of the particle velocity
(C) Wave velocity of the matter field is equal to the particle velocity
(D) None of these

8. Equation of a travelling wave is $y = 5e^{(-ax^2 - bt^2 - 2\sqrt{ab}xt)}$ when x & y is in meter and t is in sec.
$a = 25\ m^{-2}$ and $b = 9\ sec^{-2}$
(A) Travelling wave is propagated along (+) x direction.
(B) Travelling wave is propagated along (–) x direction.
(C) Speed of wave is 3/5 m/sec.
(D) Maximum displacement of particle is 5 m.

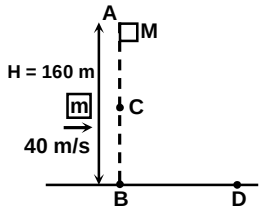
Space for Rough work

Comprehension type (Only One Option Correct)

This section contains **1 paragraph**, describing theory, experiments, data etc. Three questions relate to the one paragraph. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Questions 09 & 11

A body of mass 150 gm is dropped from point A, takes time t_1 seconds to reach the point B at the ground. Once more, the same body is dropped from point A and when body traverses half the distance, a bullet of mass 50 gm, moving horizontally with speed 40 m/s hits the falling body at point C and sticks to it. The total time of flight of the body in this case is t_2 seconds and body strikes the ground at point D. The air drag should be neglected [Take g = 10 m/s^2]. Read above passage carefully and answer the following questions.



9. The ratio of t_1/t_2 is
   (A) $\sqrt{2} : 3$ (B) $2\sqrt{2} : 3$
   (C) $3\sqrt{2} : 4$ (D) $\sqrt{2} : 4$

10. Distance between B and D is
   (A) 160 m (B) 80 m
   (C) 40 m (D) 20 m

11. Speed of the body just before striking at point D is
   (A) $5\sqrt{26}$ m/s (B) $10\sqrt{26}$ m/s
   (C) $15\sqrt{26}$ m/s (D) $20\sqrt{26}$ m/s

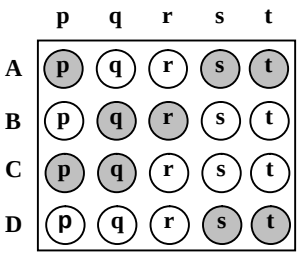
Space for Rough work

SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. An electron in a hydrogen atom makes a transition $n_1 \rightarrow n_2$, where n_1 and n_2 are the principal quantum numbers of the two states. Assume Bohr model to be valid.

Column I		Column II	
(A)	The electron emits an energy of 2.55 eV	(p)	$n_1 = 2, n_2 = 1$
(B)	Time period of the electron in the initial state is eight times that in the final state.	(q)	$n_1 = 4, n_2 = 2$
(C)	Speed of electron become two times	(r)	$n_1 = 5, n_2 = 3$
(D)	Radius of orbit of electron becomes 4.77A°	(s)	$n_1 = 6, n_2 = 3$
		(t)	$n_1 = 8, n_2 = 4$

Space for Rough work

2. A triangular wave pulse on a string, is moving in the positive x-direction with speed '*v*'. The tension in string is F and linear mass density is μ. At t = 0 shape of pulse is give by:

$$y(x,0)=\begin{cases}0 & \text{if} \quad x<-L\\ h(L+x)/L & \text{if} \quad -L<x<0\\ h(L-x)/L & \text{if} \quad 0<x<L\\ 0 & \text{if} \quad x>L\end{cases}$$

Match the column I to that in column II

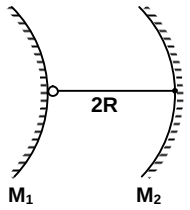
Column I		Column II	
(A)	for $-\infty < x \le vt - L$	(p)	Slope of string is negative
(B)	for $vt - L \le x \le vt$	(q)	Instantaneous power transmitted is non zero constants.
(C)	for $vt \le x \le vt + L$	(r)	Particle velocity is positive
(D)	for $vt + L \le x < \infty$	(s)	Particle velocity is negative
		(t)	Slope of string is zero

SECTION – C

(One Integer Value Correct Type)

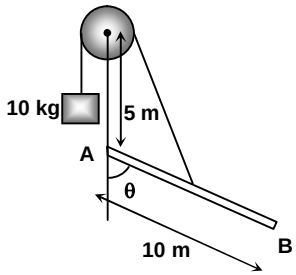
This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Two spherical mirror, one of convex and other concave, each of same radius of curvature R are arranged coaxially at a distance 2R from each other. A small circle of radius a is drawn on the convex mirror near the pole as shown in figure. The radii of 3rd image (taking the first reflection at the concave mirror, then the successive reflection at the convex mirror and finally the third successive reflection at concave mirror again) of the circle is $\left(\frac{a}{40+x}\right)$, then value of x is



Space for Rough work

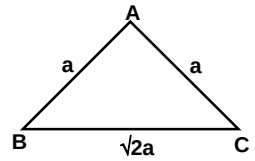
2. A uniform rod AB of mass 5 kg and length 10m is hinged at A. A block of mass 10 kg is hanging with a string and other end of the string is attached to a middle point of the rod passes over a smooth pulley as shown in the figure. Find the value of $\cos(\theta/2)$.



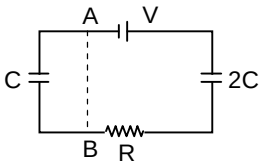
3. Pressure variation due a sonic wave propagating along positive x-direction is given by the equation $\Delta P(x) = \Delta P_0 \sin\left[2\pi(\nu t - x)\right]$. One end of an open organ pipe is closed by a cap and held at rest with its axis parallel to x-axis and the open end at the origin. The minimum value of frictional force (in Newton) between the cap and the pipe so that pipe will remain closed, is
(Given $\Delta P_0 = 2 \times 10^3$ N/m^2 and cross sectional area of the tube is equal to 5×10^{-4} m^2)

4. Two sound waves of frequencies 100 Hz and 102 Hz and having same amplitude 'A' are interfering. A stationary detector, which can detect waves of amplitude greater than or equal to A, In a given time interval of 12 seconds, find the total duration in which detector is active.

5. ABC is an isosceles triangular sheet of negligible thickness, made of isotropic material. Mass of the sheet is m. Its moment of inertia about axes perpendicular to its plane and passing through the points A and B are I_1 and I_2 respectively. [AB = AC = a and BC = $\sqrt{2}$a]. Find I_2/I_1.



6. If the heat dissipated in the circuit after a zero resistance wire is connected across the pairs A & B, is $\frac{2CV^2}{n}$, the value of n is



Space for Rough work

Chemistry

PART – II

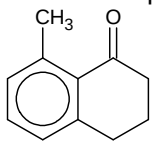
SECTION – A

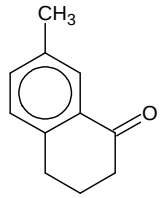
(Only One Option Correct Type)

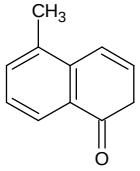
This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

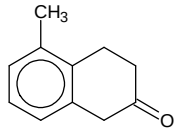
1. Calcium cyanamide is treated with ethyl chloride to give a product which on acidic hydrolysis followed by heating gives another product (A). Product (A) is treated with excess of CH_3I followed by heating with moist Ag_2O gives the product (B). Product (B) on heating strongly gives (C) which is
   (A) Ethanol (B) Ethene
   (C) Isobutene (D) Tetra allyl ammonium iodide

2. Toluene on heating with succinic anhydride in the presence of anhydrous $AlCl_3$ followed by reduction with Zn(Hg)/conc. HCl gives a product which on heating with thionyl chloride followed by heating with anhydrous $AlCl_3$ in CS_2 gives dominantly the ultimate product (P). Hence ultimate product (P) is

(A) 

(B) 

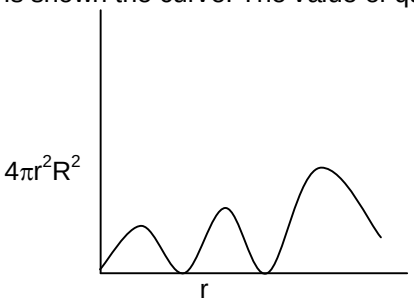
(C) 

(D) 

Space for Rough work

3. At cinema hall has equidistant rows 1m apart the length of cinema hall is 287 m and it has 287 rows. From one side of cinema hall, laughing gas N_2O is released and from the other side, weeping gas ($C_6H_5COCH_2Cl$) is released. In which rows, spectators will be laughing and weeping simultaneously?
(A) 100th row from weeping gas side
(B) 100th row from laughing gas side
(C) 187th row from the weeping gas side
(D) 185th row from the laughing gas side

4. An electron is in excited state in hydrogen atom. The orbital, in which electron is present, is shown the curve. The value of quantum numbers of electron in this exited state are



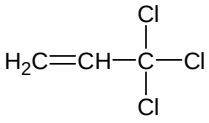
(A) n = 2, $\ell = 0$, m = 0, s = +1/2 or –1/2
(B) n = 2, $\ell = 1$, m = 0, s = +1/2 or –1/2
(C) n = 3, $\ell = 0$, m = 0, s = +1/2 or –1/2
(D) n = 3, $\ell = 0$, m = 0, s = +1/2 or –1/2

Space for Rough work

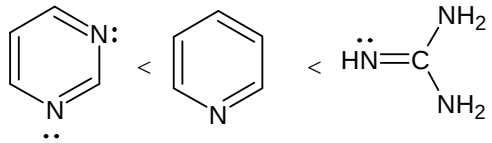
(One or More than One Options Correct Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

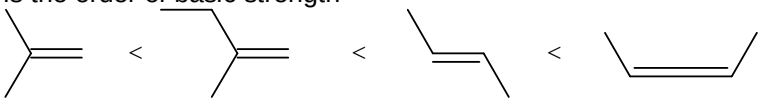
5. Identify the correct statement?

(A) 

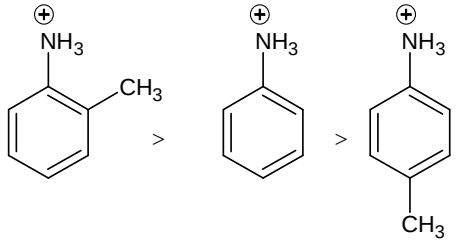
has greater ionization potential due to reverse hyper conjugation and –I effect.

(B) 

is the order of basic strength

(C) 

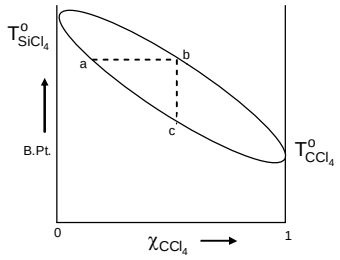
is the order of heat of combustion of the compounds.

(D) 

is the correct order of acidic strength

Space for Rough work

6. The diagram given below depicts the boiling point as the function of composition of the mixture of CCl_4 and $SiCl_4$ which of the following statements about the diagram is/are true?



(A) The point a represent the composition of solution and the point b that of the vapour in equilibrium.
(B) The proportion of CCl_4 in the solution is smaller than that in the vapour in equilibrium.
(C) bc represents the condensation of the vapour.
(D) The point c represents the composition of solution and the point b that of the vapour in equilibrium.

7. Assuming the bond direction to be Z-axis, which of the overlapping of atomic orbitals of two atoms A and B will result in bonding; (I)s orbital of A and P_x orbital of B, (II)s orbital of A and P_z orbital of B, ((III) P_y orbital of A and P_z orbital of B, (IV)s orbitals of A and B:
(A) I (B) II
(C) III (D) IV

8. Which of the following orders is are correct?
(A) $CH_3NH^- < CH_3O^- < CH_3CH_2^- \left(\text{increasing basicity}\right)$
(B) $CH_3C \equiv C^- < CH_3CH = CH^- < CH_3CH_2CH_2^- \left(\text{Increasing basicity}\right)$
(C) $CH_3OCH_3 < CH_3CH_2OH < CH_3CH_2OH_2^{\oplus} \left(\text{Increasing acidity}\right)$
(D)  > $H_5C_2-\ddot{O}-C_2H_5$
(Solubility in water)

Space for Rough work

Comprehension type (Only One Option Correct)

This section contains **1 paragraph**, describing theory, experiments, data etc. Three questions relate to the one paragraph. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Questions 09 & 11

$$\underset{\text{Mixture}}{M+P} \longrightarrow B + C\uparrow + H_2O + \underset{\text{Brown}}{D\uparrow} + \underset{\text{Colourless}}{E\uparrow}$$

$$C \xrightarrow{Ca(OH)_2} \underset{\text{White ppt}}{F} \xrightarrow{CO_2+H_2O} G$$

$$C \xrightarrow{KMnO_4/H^+} \text{No decolourisation}$$

$$P \xrightarrow{FeSO_4+H_2SO_4} \text{No brown ring observed}$$

$$P \xrightarrow{Na_2SO_4} \underset{\text{White ppt}}{Q} + \underset{\text{Solution}}{R} \xrightarrow{\text{Filter}} \xrightarrow{FeSO_4+H_2SO_4} \text{Brown ring observed}$$

$$B \longrightarrow \text{Yellow in both hot and cold condition}$$

9. What is the composition of original mixture?
   (A) $ZnCO_3 + Zn(NO_3)_2$ (B) $Na_2CO_3 + NaNO_3$
   (C) $PbCO_3 + Pb(NO_3)_2$ (D) $CuCO_3 + Cu(NO_3)_2$

10. What is F?
   (A) $ZnSO_4$ (B) $PbSO_4$
   (C) $CaSO_4$ (D) $CaCO_3$

11. $NO + E \longrightarrow$ Gas D

   $D + H_2O \longrightarrow$ Product

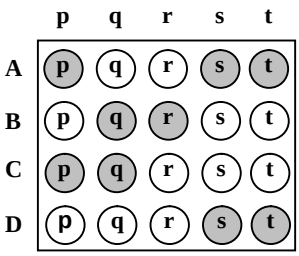
   (A) HNO_3 (B) $HNO_2 + HNO_3$
   (C) $N_2O_5 + N_2O_3$ (D) HNO_2

Space for Rough work

SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*



If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Match column I with column II such that reagent present in column II cannot be used for distinguishing pair given in column I.

Column – I		Column – II	
(A)	Phenol and benzene	(p)	Br_2 water
(B)	CH_3CHO and $(Me_3C)_2CO$	(q)	Na
(C)	HCOOH and CH_3CHO	(r)	Na_2CO_3
(D)	Phenol and ethanol	(s)	$[Ag(NH_3)_2]$ OH
		(t)	$NaHCO_3$

2. Match column (I) with column (II).

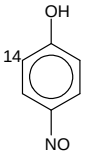
Column – I		Column – II	
(A)	$[CuCl_5]^{3-}$	(p)	Orbital involvd is $d_{x^2-y^2}$
(B)	$[Cu(NH_3)_4]^{2+}$	(q)	Orbital involved is d_{z^2}
(C)	$[FeF_6]^{3-}$	(r)	Orbital involved is d_{xy}
(D)	$[Ni(CO)_4]$	(s)	Tetrahedral
		(t)	Square planner

Space for Rough work

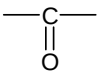
SECTION – C

(One Integer Value Correct Type)

This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. How many tautomeric structures are possible for the compound 

which have got a



group.

2. If a solid A^+B^- having ZnS structure is heated so that the ions along two of the axis passing through the face centre particles are lost and bivalent ion (Z) enters here to maintain the electrical neutrality, so that the new formula unit becomes $A_xB_yZ_{c,}$ report the value of x + y + c.

3. The area and circumference of nth orbit of a H atom are 549 $\overset{o}{A}^2$ and 83.05 $\overset{o}{A}$ respectively. The maximum number of lines that can be produced when the electron falls from this orbit to 2th orbit is

4. Calculate the $\left(\frac{pH}{1.428}\right)$ at equilibrium point when a solution of 10^{-6} M acetic acid treated with a solution of 10^{-6} NaOH, K_a for acid = 1.9×10^{-5} (Answer given in whole integer).

5. Dissolved O_2 in water is determined by using a redox reaction:

$$2Mn^{+2}(aq) + 4OH^-(aq) + O_2(g) \downarrow$$

$$2MnO_2(s) + 2H_2O(\ell)$$

How many equivalent of O_2 will be required to react with 1 mole of Mn^{+2}?

6. At high temperature, the following equilibrium exist in a mixture of carbon, oxygen and their compound:

$C(s) + O_2(g) \rightleftharpoons CO_2(g) \quad Keq_1 = 10^{-2}$

$2C(s) + O_2(g) \rightleftharpoons 2CO(g) \quad Keq_2 = 10^{-3}$

$C(s) + CO_2(g) \rightleftharpoons 2CO(g) \quad Keq_3 = X$

$2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g) \quad Keq_4 = Y$

What will be the ratio of X and Y?

Space for Rough work

Mathematics

PART – III

SECTION – A

(Only One Option Correct Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. If a, b, c are three distinct non-zero complex numbers such that |a| = |b| = |c| and the equation
$az^2 + bz + c = 0$ has a root whose modulus is 1 than
(A) $b^2 = ac$ (B) $c^2 = ab$
(C) $a^2 = bc$ (D) none of these

2. If f: [0, 1] $\to$ [0, ∞) is differentiable function with decreasing first derivative, such that f(0) = 0 and f′(x) > 0 then
(A) $\int_0^1 \frac{dx}{f^2(x)+1} > \frac{f(1)}{f'(1)}$ (B) $\int_0^1 \frac{dx}{f^2(x)+1} < \frac{f(1)}{f'(1)}$
(C) $\int_0^1 \frac{dx}{f^2(x)+1} > \frac{\tan^{-1}(f(1))}{f'(1)}$ (D) $\int_0^1 \frac{dx}{f^2(x)+1} = \frac{f(1)}{f'(1)}$

3. Let P(x) is polynomial of degree n, having n real and distinct non zero roots. Q(x) is another polynomial of degree m such that Q′(x) = 0 has at least one real root not equal to any of the root of P(x) = 0. If n, m > 2 then which of the following statements is correct
(I) : $x^2P''(x) + 3xP'(x) + P(x) = 0$ has n real roots
(II) : $P'(x)Q'(x) + P(x)Q''(x) = 0$ has at least one real root
(A) only (I) (B) only (II)
(C) both (I) and (II) (D) neither (I) nor (II)

4. Let a $\in$ [0, 4]. The maximum area bounded by the curves y = 1 – |x – 1| and y = |2x – a| is
(A) $\frac{1}{3}$ (B) $\frac{1}{4}$
(C) $\frac{1}{2}$ (D) none of these

Space for rough work

(One or More than One Options Correct Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

5. Let f, g and h are function differentiable on some open interval around 0 and satisfy the equations $f' = 2f^2gh + \frac{1}{gh}, f(0) = 1$, $g' = fg^2h + \frac{4}{fh}, g(0) = 1$ and $h' = 3fgh^2 + \frac{1}{fg}, h(0) = 1$.
The function f is given by

(A) $f(x) = 2^{-1/12}\left(\frac{\sin\left(6x+\frac{\pi}{4}\right)}{\cos^2\left(6x+\frac{\pi}{4}\right)}\right)^{1/6}$ (B) $f(x) = 2^{-1/6}\left(\frac{\sin\left(6x+\frac{\pi}{4}\right)}{\cos^2\left(6x+\frac{\pi}{4}\right)}\right)^{1/12}$

(C) $f(x) = (\sec 12x)^{1/12}(\sec 12x + \tan 12x)^{1/4}$ (D) $f(x) = (\sec 12x)^{1/4}(\sec 12x + \tan 12x)^{1/12}$

6. Let PA and PB are two tangents drawn from point P, outside the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, (a > b). If H_1 and H_2 are reflections of F_1 and F_2 respectively in the tangents PA and PB, then which of the following option(s) are correct
(A) $\angle PH_2F_1 = \angle PF_2H_1$
(B) mid-points of line joining F_1H_1 lies on a circle of radius a
(C) ΔPF_2H_1 and ΔPF_1H_2 are similar
(D) ΔPF_2H_1 and ΔPF_1H_2 are congruent

7. Let f, g and h are quadratic polynomials such that f(x) f(x + 1) = g(h(x)). If α_1, β_1 are roots of
f(x) = 0, α_2, β_2 are roots of g(x) = 0 and α_3, β_3 are roots of h(x) = 0 then which of the following is/are true?
(A) $\alpha_1 + \beta_1 = \alpha_2 + \beta_2$ (B) $\alpha_1 + \beta_1 = \alpha_3 + \beta_3$
(C) $\alpha_1\beta_1 = \alpha_2\beta_2$ (D) $\alpha_1\beta_1 = \alpha_3\beta_3$

8. For any real number $\lambda \geq 1$. let $f(\lambda)$ denote the real solution of the equation $x(1 + \ln x) = \lambda$ then
(A) $f(\lambda)$ is strictly increasing function in its domain
(B) $\lim_{\lambda\to\infty} f(\lambda) = \infty$
(C) if $h(x) = x(1 + \ln x)$, $h: [1, \infty) \to [1, \infty)$ then $f(\lambda)$ is inverse function of h(x)
(D) $\lim_{\lambda\to\infty} \frac{f(\lambda)\ln\lambda}{\lambda} = 1$

Space for rough work

Comprehension type (Only One Option Correct)

This section contains **1 paragraph**, describing theory, experiments, data etc. Three questions relate to the one paragraph. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Questions 09 & 11

Read the following write up carefully and answer the following questions:

Dot product of two vectors $\vec{A}=(a_1,a_2,a_3)$ and $\vec{B}=(b_1,b_2,b_3)$ is defined as $\vec{A}\cdot\vec{B}=a_1b_1+a_2b_2+a_3b_3$. Also, $\vec{A}\cdot\vec{B}=a_1b_1+a_2b_2$ for $\vec{A}=(a_1,a_2)$ and $\vec{B}=(b_1,b_2)$. Using these definition of dot product and algebra it can easily be proved that following equations/in-equations holds true

(I) $\vec{A}\cdot\vec{B}=\vec{B}\cdot\vec{A}$ (II) $\vec{A}\left(\vec{B}+\vec{C}\right)=\vec{A}\cdot\vec{B}+\vec{A}\cdot\vec{C}$ (III) $c\left(\vec{A}\cdot\vec{B}\right)=\left(c\vec{A}\right)\cdot\vec{B}=\vec{A}\cdot\left(c\vec{B}\right)$

9. If the dot product of two vectors $\vec{A}=(a_1,a_2)$ and $\vec{B}=(b_1,b_2)$ is defined as $\vec{A}\cdot\vec{B}=2a_1b_1+a_2b_2+a_1b_2+a_2b_1$. And other algebraic operations vectors remains as usual then

(A) (II) holds true
(B) (II) and (III) do not hold true
(C) all of (I), (II), (III) do not hold true
(D) none of these

10. If the dot product of two vectors $\vec{A}=(a_1,a_2)$ and $\vec{B}=(b_1,b_2)$ is defined as $\vec{A}\cdot\vec{B}=\sqrt{a_1b_1}+\sqrt{a_2b_2}+2\sqrt{a_2b_1}+2\sqrt{a_1b_2}$ and modulus of $\vec{A}$ is defined as $\left|\vec{A}\right|=a_1+a_2$, then which of the following is always true for a_i, $b_i>0$, $\forall$ i = 1, 2, 3

(A) $\left|\vec{A}\cdot\vec{B}\right|=\left|\vec{A}\right|\cdot\left|\vec{B}\right|$
(B) $\left|\vec{A}\cdot\vec{B}\right|>\frac{3}{2}\left|\vec{A}\right|+\left|\vec{B}\right|$
(C) $\left|\vec{A}\cdot\vec{B}\right|\le\frac{3}{2}\left(\left|\vec{A}\right|+\left|\vec{B}\right|\right)$
(D) $\left|\vec{A}\cdot\vec{B}\right|=\frac{3}{2}\left|\vec{A}\right|$

11. If however dot product of two vectors $\vec{A}=(a_1,a_2,a_3)$ and $\vec{B}=(b_1,b_2,b_3)$ in defined as $\vec{A}\cdot\vec{B}=\left|\sum_{i=1}^{3}a_ib_i\right|$, while all other algebraic operations remains as usual, then

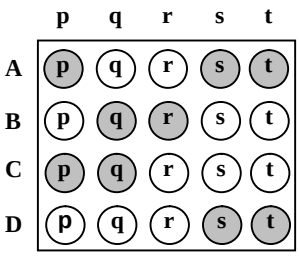
(A) (I) is not true
(B) (II) is not true
(C) all (I), (II), (III) are true
(D) (II) and (III) are true

Space for rough work

SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*



If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Parabola $y^2 = 4ax$ is given. Three normals are drawn from any point (N) inside the parabola. Tangents drawn at the intersection points (P, Q, R) of parabola and normals form an equilateral triangle (ΔABC). Then match the following Column-I with Column-II

Column – I		Column – II	
(A)	Circumcentre of ΔABC	(p)	$\left(3a, 2a\sqrt{3}\right)$
(B)	Co-ordinates of N	(q)	(5a, 0)
(C)	Co-ordinate of a point at a distance 'a' along negative x-axis from one of the intersection point (P, Q, R) is	(r)	(0, 0)
(D)	Co-ordinates of intersection points	(s)	(–a, 0)
		(t)	$\left(3a, -2a\sqrt{3}\right)$

2. Match the following Column-I with Column-II

Column – I		Column – II	
(A)	If $f(x) + f''(x) = -x\, g(x)f'(x)$ and $g(x) > 0\ x \in R$ then $(f(x))^2 + [f'(x)]^2$ has k as its maxima. Then k is	(p)	0
(B)	If $f^2(x) + f(x) = g(x)$ and g(x) is always increasing then the minimum value that f(x) can attain so that f(x) also increasing is k. Then (2(k + 1)) is	(q)	1
(C)	If $g(x) = f(x)\int_0^x f(t)dt$ is non-increasing function and f(x) is continuous in R then f(1) is	(r)	2
(D)	If f(x) is differentiable function and $f(x) = x^2 + \int_0^x e^{-t} f(x-t)dt$ then $f(-1) = \frac{k}{3}$ then k is	(s)	–1
		(t)	3

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. If $\lim\limits_{r\to\infty} \dfrac{r^C \int_0^{\pi/2} x^r \sin x\,dx}{\int_0^{\pi/2} x^r \cos x\,dx} = L$, then value of $\pi L - C$ is equal to _____ ($C \in R, L > 0$)

2. Let $f: R \to R$, satisfy the equation $(x - y)f(x + y) - (x + y)f(x - y) = 4xy(x^2 - y^2)\ \forall\ x, y \in R$. If $f(1) = -2$, then the absolute value of difference between maximum and minimum value of $f(x)$ on the interval $x \in \left[-\sqrt{3}, \sqrt{3}\right]$ is _____

3. The number of all the pairs of natural numbers (x, y) which satisfy the equation $2x(xy - 2y - 3) = (x + y)(3x + y)$ is _____

4. The number of distinct solutions $x \in [0, \pi]$ which satisfy the equation $8 \cos x \cos 4x \cos 5x = 1$ is k then $\dfrac{k}{2}$ is equal to _____

5. n biased coins, with m^{th} coin having probability of throwing head equal to $\dfrac{1}{(2m+1)}$ ($m = 1, 2, \ldots, n$), are tossed once. The probability of getting an odd numbers of heads, if results for each coin are independent, is $\dfrac{\alpha n}{\beta n + \gamma}$ then $\alpha + \beta + \gamma$ is equal to _____

6. Consider the parabola $y = ax - bx^2$. If the least positive value of a for which there exist $\alpha, \beta \in R - \{0\}$ such that both the point (α, β) and (β, α) lies on the given parabola is k then [k] is equal to _____

Space for rough work